

Project Design Phase

FitBuddy AI — Project Deliverable

Convert the analyzed requirements into a clear problem-solution fit, proposed solution, and high-level architecture.

Deliverables in this single phase document

- Problem-Solution Fit
- Proposed Solution
- Solution Architecture

Evidence status — Architecture is derived from the documented file roles and README; exact internal implementation should be confirmed in source code.

3.1 Problem-Solution Fit

The documented problem is the need for a simple, structured wellness-planning workflow. FitBuddy AI addresses it with profile-driven 7-day plan generation, a nutrition/recovery tip, and feedback-based revision inside a browser application.

3.2 Proposed Solution

- Responsive browser UI for profile entry and plan review.
- FastAPI backend for browser routes and JSON APIs.
- Pydantic-based validation for incoming data.
- Gemini-powered generation when enabled, with a local demo mode for running without Gemini.
- SQLite + SQLAlchemy for persistent local storage.
- Feedback route and revised-plan behavior.
- Automated tests using pytest and httpx.

3.3 Solution Architecture

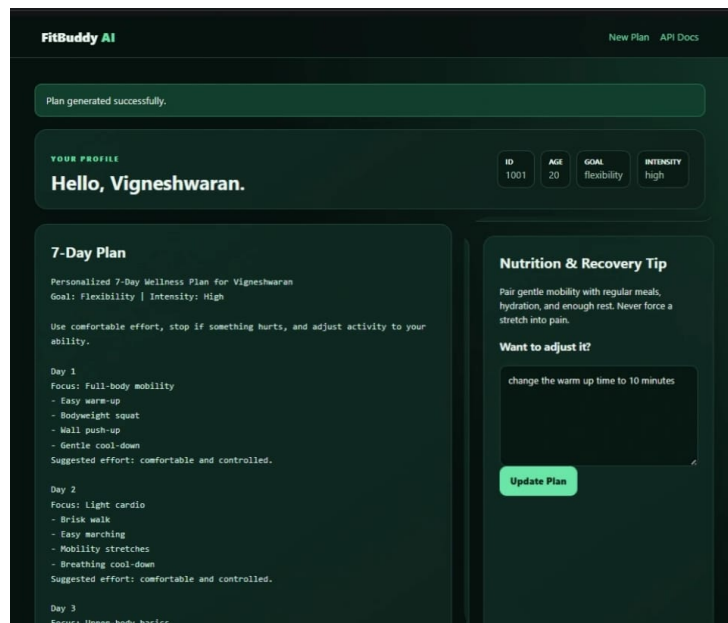
- Presentation layer: Jinja2 templates and responsive HTML/CSS.
- Application/API layer: FastAPI routes and request/response handling.
- Validation/configuration layer: Pydantic and pydantic-settings.
- Generation layer: Gemini integration in gemini_generator.py, with configurable models.
- Persistence layer: SQLite database accessed through SQLAlchemy.
- Revision layer: feedback processing and updated-plan behavior.

3.4 Project Structure Mapping

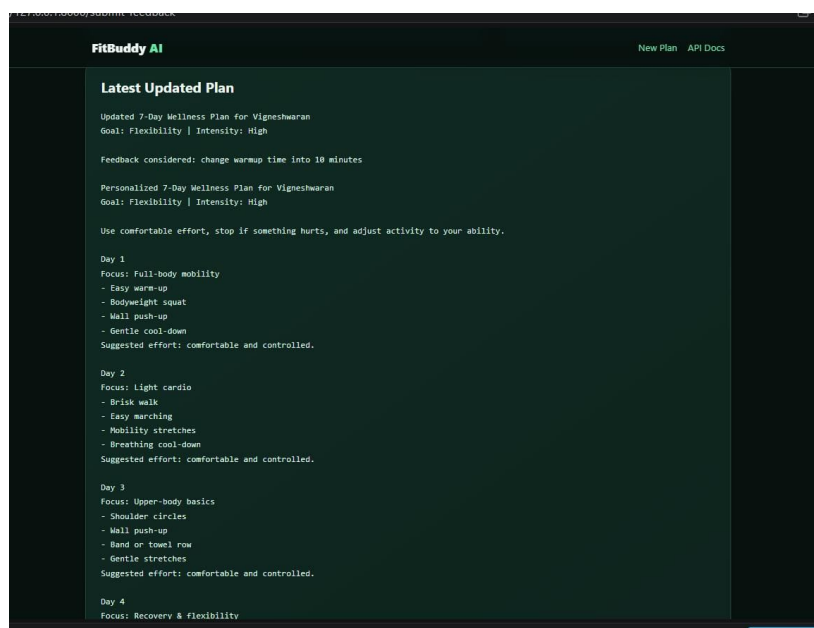
The supplied README identifies app/config.py, database.py, gemini_generator.py, main.py, models.py, routes.py, schemas.py, and updated_plan.py, plus templates, static assets, tests, and configuration files.

Area	Key files
Backend	app/main.py, routes.py, schemas.py
Data	app/database.py, models.py
AI generation	app/gemini_generator.py, updated_plan.py
UI	templates/index.html, result.html, all_users.html; static/style.css
Testing	tests/test_app.py

Project Evidence / Screenshots



Screenshot evidence: generated result page.



Screenshot evidence: updated plan after feedback.

Source note: This phase document was converted from the supplied FitBuddy AI project documentation, README, configuration/dependency information, and supplied project screenshots. Where the source did not provide a measured result, named team member, or actual date, the document does not present one as a completed fact.